

Arjun Nair

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EDUCATION

Stanford University

Sept. 2025 – June 2029

Bachelor of Science in Computer Science — GPA - 3.75

- **Relevant Coursework:** Operating Systems, Introduction to Embedded Systems, Circuits, Computer Systems, Data Structures & Algorithms, Probability for Computer Science

EXPERIENCE

Embedded Software Engineering Intern, Lotus

June 2026 – Aug. 2026

Assistive technology startup developing wearable hardware for accessible device control

- Developed low-level embedded C firmware and **device drivers** for an assistive wearable using Zephyr RTOS and the nRF Connect SDK, supporting **hardware bring-up and driver debugging**.
- Built IR interference detection to detect 3 or more failed decodes within 10-second windows, distinguishing valid transmissions from interference and triggering a user alert when persistent IR noise was detected.
- Migrated power-management firmware from the TI BQ25180 to BQ25188, adapting **register-level configuration** for **4.5 V input regulation** and **1.5 A battery protection** while updating interrupt behavior for the new PMIC.

Remote Sensing Research Intern, NASA

June 2024 – Oct. 2024

Earth Systems Explorer Team, UT Austin Center for Space Research

- Built a remote-sensing pipeline using Landsat time-series data across U.S. agrivoltaic sites, aligning solar installation dates with 25-year vegetation histories to assess pre- and post-installation vegetation change.
- Quantified post-installation vegetation impacts, measuring an **18% decrease in Normalized Difference Vegetation Index (NDVI)** and **54% reduction in Tasseled Cap Greenness**.
- Presented findings at the **American Geophysical Union (AGU) Annual Meeting**.

Student Researcher, MIT Lincoln Laboratory

July 2024 – Aug. 2024

Remote Sensing for Disaster Response

- Developed and evaluated CNN-based models for hurricane damage assessment from aerial imagery, achieving **78% classification accuracy** across **5 damage levels** and analyzing confusion matrices to identify model failure modes.
- **Tested and retrained damage classifiers** on hurricane imagery to improve infrastructure assessment, turning model predictions into geospatial damage analysis for post-disaster response and search-and-rescue planning.

PROJECTS

Real-Time ARM Motor Controller | C, ARM Cortex-M3, QEMU, GDB

2026

- Built a bare-metal ARM Cortex-M3 motor-control system with zero OS/runtime dependencies, using **memory-mapped peripheral drivers** and **hardware timers** to sustain **20 kHz PWM**.
- Designed a **100 Hz integer PID speed controller** with anti-windup for bidirectional **±3,000 RPM** control, achieving **less than 1% steady-state error** and **2.1 s settling time** with **zero missed control deadlines**.
- Implemented **latched overspeed** and **emergency-stop protection** with deterministic recovery, then validated safe operation under full direction reversals and hardware faults through runtime **UART telemetry**.

Distributed Key-Value Store | C++, gRPC, Protobuf, Raft

2026

- Built a distributed key-value store to explore durability and fault tolerance under machine failures, designing persistent storage with write-ahead logging and crash recovery.
- Implemented multi-node replication over gRPC and Protocol Buffers, using **Raft consensus** to coordinate leader election and majority-committed writes while tolerating single-node failures.
- Benchmarked a 3-node cluster at **5.7K writes/sec**, **22 ms average latency**, **42 ms p99 latency**, and **around 202 ms leader failover** with **zero failed requests**.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, HTML/CSS

Systems & Frameworks: Zephyr RTOS, Linux, POSIX, TCP/IP, gRPC, Protocol Buffers

Developer Tools: GDB, QEMU, nRF Connect SDK, Git, CMake, Docker, wrk